

CY 002: Preparatory Chemistry - 2

Course Code	CY 002
Course Name	Preparatory Chemistry - 2
L-T-P-C	3-1-0-4
Intended for	Preparatory Students
Distribution	Core
Approval	50th BoA (Jul 2023)

Course Contents

Physical Chemistry

- **Quantum Mechanics:** Construction of Hamiltonian operator; solution of $H\Psi=E\Psi$ for particle in a 1-d box; normalization and orthogonality of Ψ , nodes in excited states, and calculation of average values like $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$ and $\langle p^2 \rangle$, demonstration of the uncertainty product inequality, $\Delta x \Delta p \geq h/4\pi$, discussion on uncertainty principle.
- **The H atom problem:** Hamiltonian in Cartesian and Polar Coordinates; separation of radial and angular parts; emergence of magnetic quantum number; mathematical forms of orbital functions (ns and np) and degeneracy, shapes of orbitals (s,p).
- **Spectroscopy and Photochemistry:** Einstein's Law, primary photophysical processes; potential energy diagram; Frank-Condon Principle; fluorescence and phosphorescence; photochemical reactions, quantum yield; photosensitisation; photochemical equilibrium; dimerization of anthracene.
- **Alkali-Metal Spectra (S, P, D, F series):** its origin, multiplicity of spectral lines, idea of spin quantum number; physical idea of spin-orbit coupling, rotational (rigid rotator model) and vibrational (harmonic oscillator model) spectra of diatomics; frequency expressions, applications to estimate molecular parameters, idea of $n \rightarrow \pi^*$ and $\pi \rightarrow \pi^*$ electronic spectra; conjugated polyenes and 1-d box model.
- **Dipole moment and Intermolecular Forces:** Induced and orientation polarisation; Debye and Clausius Mossotti equations (with derivations), dipole-dipole, dipole-induced dipole and van der Waals interactions in molecules, realistic intermolecular potential energy diagrams.

Inorganic Chemistry

- **Inorganic Chemistry:** Chemistry of d-block elements, crystal field theory, magnetism in transition metal compounds, valence bond theory for prediction of molecular geometry, magnetic properties, metal-carbonyl chemistry, important elements of catalysis by transition metal compounds, chemistry of f-block elements.

Organic Chemistry

- **Organic Chemistry:** Functional group inter-conversions, concept of stereochemistry concept of aromaticity, aromatic electrophilic and nucleophilic substitution reactions.

Text Books

1. J. D. Lee, Concise **Inorganic Chemistry**, 5th Edition, ELBS, 1996.
2. R. T. Morrison and R.N. Boyd, **Organic Chemistry**, 5th Edition, Printice Hall of India Pvt. Ltd. 1990.
3. G. Solomons and C. Fryhle, **Organic Chemistry**, John Wiley and Sons (Asia) Pte Ltd.
4. P. W. Atkins, **Molecular Quantum Mechanics**, Oxford University Press, 1999.